



HỒ KHẮC HUY

Senior Software Engineer · Distributed Systems, Reliability & Applied AI

● AVAILABLE IMMEDIATELY

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1.64M scans/s
PromptShield · LLM01

102k evals/s
AgentToolGuard · 5.3µs

6,661 req/s
HookSentinel · Ingest

847k writes/s
CrashProofWAL · 15ms

PR #131 Merged
Upstream Contributor

TECHNICAL ARSENAL

LANGUAGES

Python, TypeScript, JavaScript (ES6+), SQL, Bash

APPLIED AI & AGENTS

OWASP LLM01, Heuristic Jailbreak Defense, Agent Tool-Call Firewalls, Cycle Detection, Semantic Caching, Grounded RAG Evaluation

BACKEND & DISTRIBUTED

FastAPI, Node.js, asyncio, REST APIs, HMAC-SHA256 Webhooks, Idempotency Ledgers, DLQ, Circuit Breakers, Kahn's DAG

STORAGE & INFRASTRUCTURE

Write-Ahead Logs (CRC32, Merkle Proofs), SQLite WAL, PostgreSQL, Docker, GitHub Actions, Zod, pytest, Playwright

ENGINEERING FOCUS

- Deterministic AI agent safety
- Cryptographic webhook defense
- Crash-resilient write pipelines
- Topological DAG orchestration
- Sub-millisecond p99 latency

EDUCATION

RMIT University Vietnam

Bachelor of Business · Systems Focus
2021 – 2025 · Ho Chi Minh City
Applied software systems, distributed architectures & API engineering.

UPSTREAM CONTRIBUTOR

python-docx-ng

Merged Upstream PR #131

Resolved XPath quotation bug; authored regression test matrix.

VERIFIED SYSTEMS & APPLIED AI ARCHITECTURES

Benchmarked · Open Source · Live Sandboxes

PromptShield — Enterprise LLM Security Proxy & Fallback Router

Sep 2026

Author · 7 Tests · [Live Radar](#) · [TypeScript](#) [Node.js](#) [OWASP LLM01](#) [Circuit Breaker](#)

- **Heuristic Jailbreak Scanner:** Engineered zero-dependency threat detector intercepting DAN jailbreaks and system-prompt overrides at **1.64M scans/s** (0.4µs latency); integrated token-bucket rate limiter.
- **Semantic Cache & Fallback:** Built deterministic SHA-256 caching delivering **690k reads/s** (<1µs hits, 100% token savings); automated OpenAI ↔ Anthropic circuit breaker fallback.

AgentToolGuard — AI Agent Tool-Call Firewall & Loop Interceptor

Sep 2026

Author · 10 Tests · [Python](#) [FastAPI](#) [Schema Firewall](#) [Cycle Detection](#)

- **Sub-Millisecond Schema Boundary:** Engineered deterministic tool-calling validator evaluating payloads in **5.33µs (102,516 evals/s)**; deflected 100% of injected SQL/shell payloads at **435,640 attacks/s**.
- **Cycle & Blast Radius Interceptor:** Built sliding-window hash chain intercepting agent thrashing and oscillating loops; enforced cryptographic token authentication for destructive mutations.

HookSentinel — Ingestion & Dead-Letter Broker

Sep 2026

Author · 78 Tests · [Live Demo](#) · [FastAPI](#) [Python](#) [SQLite WAL](#) [Docker](#)

- **Empirical Ingest Benchmark:** Multi-provider constant-time HMAC verification (Stripe, GitHub, Shopify) achieving **6,661 req/s** with sub-millisecond p99 latency (0.58ms).
- **Replay Defense:** Intercepted 100% of concurrent replay bursts (11,904 checks/s) via deterministic idempotency locks ("provider:sha256(body)"); isolated poisoned deliveries to DLQ.

CrashProofWAL — Crash-Resilient Write-Ahead Log Store

Sep 2026

Author · 5 Tests · [Live Demo](#) · [Python](#) [Binary Protocol](#) [CRC32](#) [Merkle Tree](#)

- **Binary Append-Only Durability:** Engineered binary storage engine with hardware CRC32 checksums processing **847,449 writes/s (72.4 MB/s)** and in-memory index serving **2.7M reads/s**.
- **Surgical Crash Recovery:** Built automatic corrupted-tail truncation recovering from `SIGKILL` in **15.03ms** with 0 bytes committed data lost; integrated SHA-256 Merkle tree state proofs.

ResilientDAG — Fault-Tolerant DAG Data Pipeline

Sep 2026

Author · 10 Tests · [Live Demo](#) · [TypeScript](#) [Node.js](#) [Kahn DAG](#) [Zod](#)

- **Topological DAG Solver:** Implemented Kahn's algorithm resolving execution tiers at **179,421 graphs/s** (4.2µs p50) with 100% circular deadlock interception (115k checks/s).

COMMERCIAL & PRODUCTION ENGINEERING

Production Deployments & Client Impact

Independent Software Engineering Practice

Aug 2023 – Present

Senior Software Engineer · Ho Chi Minh City, Vietnam · [Node.js](#) [Express](#) [PostgreSQL](#)

- **60-Second Lead Capture Engine:** Architected durable lead ingestion service (Node.js/Express, rate limiting, Telegram webhook alerts, write-ahead failover logging) delivering sub-minute response times and zero lead loss.
- **Automotive Workforce Attendance Platform:** Built operations web application for an automotive manufacturing plant in Vietnam, translating complex shift scheduling and punch-in rules into validated relational data models.
- **ERP Data Pipeline & Reporting:** Engineered automated ETL reconciliation scripts extracting transaction records from Odoo ERP and spreadsheets into audited multi-format reports (Excel, HTML, PDF).

Support Readiness AI Workbench — E-Commerce LLM Evaluator

2026

Author · 11 Tests · [Live Evidence](#) · [Next.js](#) [React](#) [TypeScript](#) [LLM Evaluation](#)

- **Deterministic AI Support Guardrails:** Engineered evaluation harness classifying customer support tickets into review-ready drafts vs. mandatory human escalations; prevented hallucinations and policy violations across 15 synthetic cases.